104

Plasma Exchange c010

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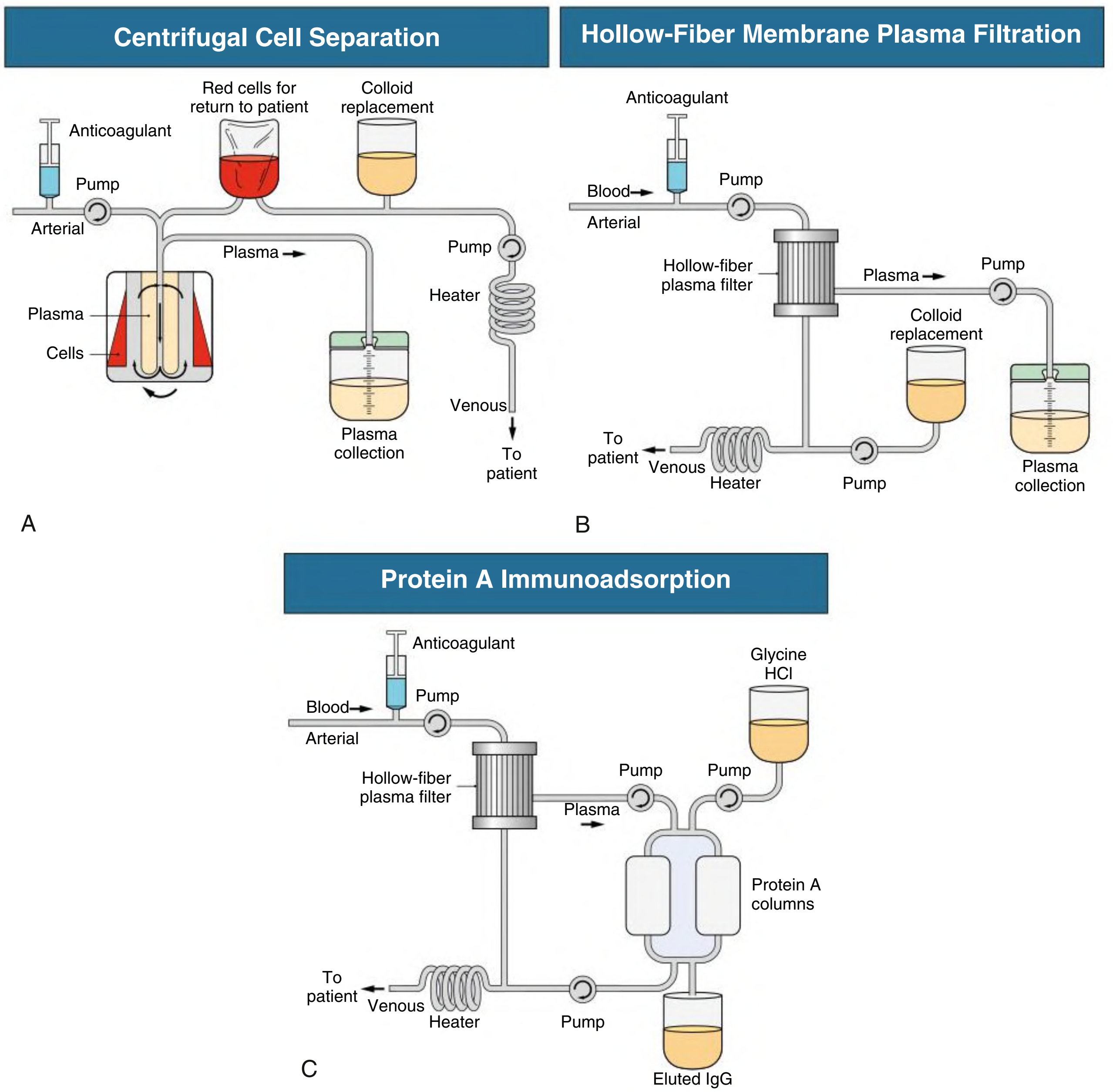
p0010 Over the past 10 years, it has become clearer how best to use plasma exchange (plasmapheresis) in the management of kidney disease, but the quality of published data remains relatively poor, with less than 1% of relevant literature in the form of randomized controlled trials (RCTs). Plasma exchange came into widespread clinical use after early reports of beneficial effects in Goodpasture disease in the mid-1970s. It is used to remove many large-molecular-weight substances from plasma, including pathogenic antibodies, cryoglobulins, and lipoproteins. Newer techniques allow more selective removal of plasma components, such as double-filtration plasma exchange, cryofiltration, and immunoadsorption with or without immobilized ligands. The most common kidney indications are thrombotic thrombocytopenic purpura/hemolytic-uremic syndrome (TTP/HUS), kidney transplantation (antibody-mediated rejection and for desensitization either for anti-ABO blood group antibodies or anti–human leukocyte antigen [HLA] antibodies), antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis, cryoglobulinemia, recurrent focal segmental glomerulosclerosis (FSGS), and anti–glomerular basement membrane (anti-GBM) antibody disease.

s0010 TECHNIQUES

p0015 Plasma exchange can be carried out either by centrifugal cell separators or (more commonly in dialysis units) with hollow-fiber plasma filters (membrane filtration) and standard hemodialysis (HD) equipment (Figs. 104.1 and 104.2). Centrifugal devices allow withdrawal of plasma from a bowl with either synchronous or intermittent return of blood cells to the patient.[1] There is no upper limit to the molecular weight of proteins removed by this method. The bowls and circuits are single use and disposable and require relatively low blood flow rates (50–150 mL/min and can be as low as 5 mL/min for children) that can be delivered through large-bore peripheral cannulae. Platelets can be inadvertently removed by centrifugal devices, although this risk has reduced significantly with newer technologic approaches. Membrane plasma filtration uses highly permeable hollow fibers with membrane pores of 0.2 to 0.5 μm. Plasma readily passes through the membrane and the cells are simultaneously returned to the patient. All immunoglobulins will cross the membrane (immunoglobulin G [IgG] slightly more efficiently than immunoglobulin M [IgM]), and molecules up to 3 million Da are cleared. Hemolysis can occur if transmembrane pressures are too high (a rare complication). The blood flow rates required for membrane plasma filtration are higher than for centrifugal cell separators (100–300 mL/min) and require central venous access or a fistula. Membrane exchange takes slightly longer than centrifugal techniques for the same plasma volume removal. It has been suggested that the adsorptive properties of the membrane for cytokines and other biomolecules may account for some of the beneficial effects of plasma filtration. There have been occasional reports of mild adverse reactions in patients taking angiotensin-converting enzyme inhibitors when plasma is filtered with ethylene vinyl alcohol or acrylic copolymer membranes. Reuse of plasma filters is not advised because of potential risks resulting from loss of filtration capacity and to staff from cleaning procedures, but performance data do not indicate a major loss of function during routine plasma exchange. For patients with severe kidney failure, sequential HD and plasma exchange can easily be performed with plasma filtration.

Vascular access is usually achieved using standard central venous catheters, but existing arteriovenous fistulas (AVFs) can be used if available. Sometimes plasma exchange can be done using large-bore, short intravenous (IV) cannulas placed in the antecubital fossa, especially with centrifugal devices. Single-needle access using an AVF is also relatively easy to accommodate, especially for centrifugal plasma exchange, in which the blood removal and return can be asynchronous, but also for membrane filtration. Anticoagulation must be carefully managed in patients at higher bleeding risk (e.g., thrombotic microangiopathy [TMA], recent or ongoing pulmonary hemorrhage, or a recent kidney biopsy). Citrate is used for anticoagulation with centrifugal plasma exchange and heparin for membrane plasma filtration; however, citrate is superior for patients at higher bleeding risk in view of its lack of systemic anticoagulation.[1] When heparin is used, higher doses may be needed than in HD because of increased losses during the procedure (heparin is protein bound). Bolus doses of unfractionated heparin of 2000 to 5000 U are given initially and then 500 to 2000 U/h. Anticoagulant is administered prefilter. Increasingly low-molecular-weight heparin (LMWH) is also used with a single bolus dose at initiation of exchange. p0020

Both methods of plasma exchange require large volumes of colloid replacement. A single plasma volume exchange will lower plasma macromolecule levels by approximately 60%, and five exchanges over 5 to 10 days will clear 90% of the total body immunoglobulin (Fig. 104.3).[1,2] For most patients, this is achieved by removing 50 mL of plasma per kilogram body weight at each procedure (~4 L for a 75-kg person). Daily plasma exchange more rapidly depletes total body load of immunoglobulins, but there is no good evidence that intensity of exchanges has a major effect on outcomes except in patients with HUS with poor prognostic markers (see later discussion). Indeed, alternate-day exchanges are of proven efficacy in ANCA-associated diseases. Replacement solely with crystalloid is contraindicated because of the need to maintain colloid oncotic pressure. Synthetic gelatin–based plasma expanders or hydroxyethyl starch (Hespan) can be used as part of a replacement regimen but have been reported to cause a coagulopathy in patients with sepsis and have a shorter half-life than human albumin, which is the main replacement fluid. The major disadvantage of albumin solutions is the lack of clotting factors, with the potential development of depletion coagulopathy after plasma exchange. Fresh-frozen plasma (FFP) p0025



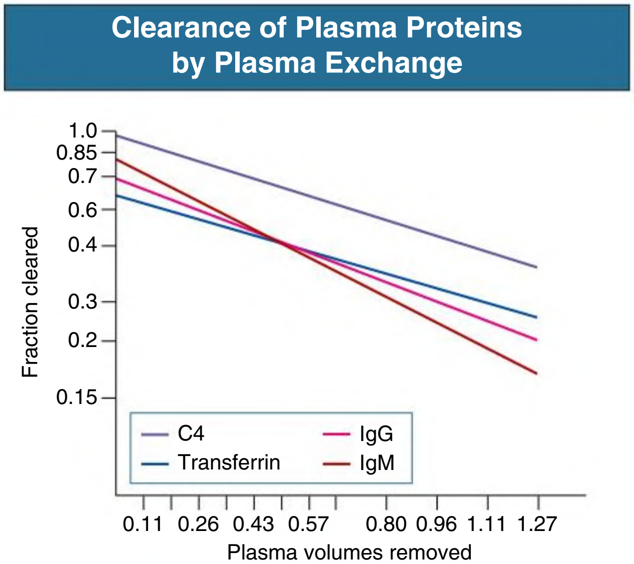
f0010 **Fig. 104.1** Plasma Exchange and Immunoadsorption Techniques. Techniques include centrifugal cell separation (A), hollow-fiber membrane plasma filtration (B), and protein A immunoadsorption (C). *IgG,* Immunoglobulin G.

should be given, usually in addition to human albumin solution, in patients at particular risk for bleeding. If partial replacement is with FFP, this should be given late during the exchange so the constituents are not removed by the ongoing plasma exchange. However, almost all the serious complications of plasma exchange (hypotension, anaphylaxis, citrate-induced paresthesia, urticaria) have been reported in patients receiving FFP rather than albumin (see later discussion).[1,3] Both human products carry a tiny risk for transmission of infectious diseases, especially viral. Standard regimens for plasma exchange are summarized in Table 104.1. Human albumin solution (4%–5%) should be used for all exchanges except in TMAs (in which plasma should provide the total exchange), and FFP should form part of the exchange when bleeding risk is high (ongoing pulmonary hemorrhage or within 48 hours of biopsy or surgery). If fibrinogen levels decrease to less than 1.25 to 1.5 g/L or prothrombin time (PT) is increased 2 to 3 seconds above normal, FFP should be administered (Table 104.2).

Double-filtration plasma exchange (or cascade filtration) uses membrane filtration to separate cells from plasma and then a secondary plasma filtration (pore size 0.01–0.03 μm) to remove plasma solutes based on molecular size. Most albumin is therefore returned to the patient, together with lower molecular weight proteins, reducing the need for replacement fluids. Cryofiltration uses a similar principle but exposes the filtrate to 4°C during the procedure, with the aim of precipitating cryoproteins. These techniques are not widely available. p0030

Selective and specific immunoadsorption techniques are available but rarely used. Protein A immunoadsorption has been used to remove immunoglobulin alone from plasma, without the need for replacement fluids and without depletion of clotting factors and complement (see Fig. 104.1C). Protein A selectively binds the Fc domains of p0035

f0015 **Fig. 104.2** A centrifugal cell separator in use for plasma exchange.



f0020 **Fig. 104.3** Clearance of Plasma Proteins by Plasma Exchange. Clearance from the intravascular compartment varies with the plasma volume exchanged and among individual proteins. *IgG,* Immunoglobulin G; *IgM,* immunoglobulin M. (Modified from Derksen RH, Schuurman HJ, Meyling FH, et al. The efficacy of plasma exchange in the removal of plasma components. *J Lab Clin Med.* 1984;104:346–354.)

immunoglobulin molecules, and the immunoadsorption columns can be repeatedly regenerated. Columns have been used for 1 year for a single patient on up to 30 occasions; however, the repeated acid stripping during regeneration reduces the efficacy of antibody binding. Selective immunoadsorption has been used to treat conditions in which autoantibodies are thought to be important and usually in place of plasma exchange (e.g., Goodpasture disease, rheumatoid arthritis, lupus, or systemic vasculitis) and to remove anti-ABO or anti–HLA antibodies in highly sensitized transplant recipients. In general, the reported efficacy has been equal to that of plasma exchange, although if used over the long term, immunoadsorption is much more cost-effective in single patients because replacement of albumin or plasma is not required. Specific ligands also have been immobilized onto columns for more specific removal of potentially pathogenic serum factors; ligands used include anti–human IgG, C1q, phenylalanine, hydrophobic amino acids, acetylcholine receptor, β-adrenoreceptor peptides, and blood group–related oligosaccharides. Immunoadsorption is not widely available because the initial column costs are high and few centers have much clinical experience. Costs of all other plasma exchange modalities are dominated by the costs of replacement albumin and plasma; equipment costs do not vary much among modalities, and all require skilled and trained nursing staff.

COMPLICATIONS s0015

The complication rate of plasma exchange is not high.[3] The Swedish registry reported no fatalities during 20,485 procedures and an overall adverse incidence rate of only 4.3% of all exchanges (0.9% for severe adverse events) of which 27% were paresthesias, 19% transient hypotension, 13% urticaria, and 8% nausea.[4] The Canadian Apheresis Registry collected data on 144,432 apheresis procedures since 1981 and reported adverse events occurring in 12% of procedures (mostly minor) and overall in 40% of patients. Severe events occurred in only 0.4% of procedures. Three deaths were probably related directly to the procedure: one from a transfusion-related acute lung injury and two from complications from central venous catheters.[5] An overall complication rate of 1.4% has been reported in more than 15,000 treatments in patients receiving albumin and 20% in patients receiving FFP.[1,3,5,6] Plasma exchange by centrifugation had a lower risk for adverse events than by filtration. p0040

Other complications directly attributable to plasma exchange include citrate-induced hypocalcemia (presenting with perioral tingling and paresthesias) and citrate-induced metabolic alkalosis. Citrate is usually present in FFP (up to 14% by volume) or is administered in the extracorporeal circuit as an anticoagulant; it binds free calcium in plasma. Symptomatic hypocalcemia can be averted by infusing 10 to 20 mL of 10% calcium gluconate during each plasma exchange. Alkalosis is rare and is caused by metabolism of citrate to bicarbonate and failure to excrete the latter in patients with kidney impairment. p0045

Plasma exchange predictably increases the risk for bleeding by depleting coagulation factors in patients receiving albumin as sole replacement colloid. PT is increased by 30%, and partial thromboplastin time by 100% after a single plasma volume exchange. Patients at risk for bleeding (pulmonary hemorrhage, postbiopsy, postoperative) should receive FFP (300–600 mL) with replacement fluids. Dilutional hypokalemia is avoided by replacing potassium according to daily blood testing. An increased incidence of infection secondary to hypogammaglobulinemia has not been confirmed in recent series.[3–6] Sepsis related to IV access is the most common infectious complication of plasma exchange. Hypotension can occur for a variety of reasons related to extracorporeal circuits, sepsis, and allergic reactions. Cascade filtration can lead to hemolysis (in up to 20% patients) but rarely necessitates transfusion. p0050

MECHANISMS OF ACTION s0020

The pathogenicity of autoantibodies in anti-GBM disease (Goodpasture disease) provided the impetus for development of plasma exchange therapy, but it is now clear that antibodies, although necessary, are not alone sufficient to cause the necrotizing glomerulonephritis (GN) in that disease. Plasma exchange, however, removes p0055 t0010 **TABLE 104.1 Practical Regimens for Plasma Exchange in Kidney Disease**

Indication	Anti-GBM Disease	Small-Vessel Vasculitis	Cryoglobulinemia	Recurrent FSGS After Transplantation	HUS/TTP
Duration of treatment	Daily, at least 14 days until anti-GBM antibodies 20% of baseline	Daily or alternate daily, 7–10 days depending on clinical response	At least 7–10 days or until clinical response	Daily for at least 10 days initially, then continuing less frequently, often for months	Daily for 7–10 days or until platelet count 80–100 $\times 10^9$/L (sometimes needed twice daily)
Exchange volume	50 mL/kg each treatment	As for anti-GBM disease	As for anti-GBM disease	As for anti-GBM disease	As for anti-GBM disease
Replacement fluid	Human albumin 5% (unless bleeding risk)	As for anti-GBM disease	As for anti-GBM disease	As for anti-GBM disease	FFP or cryo-poor FFP
Additions to replacement fluid	20 mL 10% calcium gluconate (occasionally more), 3 mL 15% KCl if not dialysis dependent, heparin 2000–5000 U bolus, then 500–2000/h (LMWH an alternative with bolus only) *or* citrate anticoagulation	As for anti-GBM disease	As for anti-GBM disease	As for anti-GBM disease	As for anti-GBM disease (may need more calcium because of increased volume of FFP-containing citrate)
Immunosuppression	See Chapter 25	See Chapter 26	See Chapter 22	See Chapter 30	See Chapter 30
Variations	FFP 5–8 mL/kg at end of exchange volume if hemorrhage risk (kidney biopsy in last 48 h, lung hemorrhage, platelets <40 $\times 10^9$/L, fibrinogen < 1.5 g/L) Immunoadsorption may be as effective	As for anti-GBM disease Immunoadsorption may be as effective	As for anti-GBM disease	As for anti-GBM disease; may need to include replacement immunoglobulins if continuing long term Immunoadsorption may be as effective	

FFP, Fresh-frozen plasma; *FSGS,* focal segmental glomerulosclerosis; *GBM,* glomerular basement membrane; *HUS,* hemolytic-uremic syndrome; *LMWH,* low-molecular-weight heparin; *TTP,* thrombotic thrombocytopenic purpura.

t0015 **TABLE 104.2 Investigations to Undertake Before Every Plasma Exchange Session**

Investigation	Action to Take
Platelet count	Use plasma exchange with caution if platelet count < 40 × 10^9/L unless disease itself associated with thrombocytopenia (e.g., aHUS/TTP), and use FFP as part of replacement fluid.
Plasma fibrinogen	If <150 mg/dL, should use FFP as part of exchange fluid.
Prothrombin time	If prolonged more than 2–3 seconds, use FFP as part of exchange fluids.
Serum calcium	If <2.1 mmol/L (<8.4 mg/dL), increase amount being provided with replacement albumin/FFP.
Serum potassium	Can be highly variable depending on changing kidney function in addition to plasma exchange treatment; adjust potassium replacement according to serum potassium.

aHUS, Atypical hemolytic-uremic syndrome; *FFP,* fresh-frozen plasma; *TTP,* thrombotic thrombocytopenic purpura.

all large-molecular-weight substances from the plasma in addition to antibodies, including complement components, immune complexes, endotoxin, lipoproteins, and von Willebrand factor (vWF) multimers. In animal studies, for example, complement depletion abrogates anti-GBM GN very effectively. Therefore, plasma exchange may have benefits in addition to clearance of autoantibodies.

p0060 The clearance of antibodies from patients is variable and depends on several factors, including the rate of equilibration of macromolecules between the intravascular and extravascular compartments. IgM antibodies are cleared more effectively than other classes of immunoglobulin because they are retained in the vascular compartment almost wholly. A rebound increase in antibody production will occur unless there is concomitant immunosuppression to prevent resynthesis.

p0065 Plasma exchange has been shown to remove immune complexes, which may have clinical significance in cryoglobulinemia and systemic lupus, and fibrinogen and complement components. There is no good evidence that removal of cytokines has any clinical significance. Plasma exchange reduces plasma viscosity, with consequent improved blood flow in the microvasculature. There is also some evidence for improvement in monocyte/macrophage function, alteration in lymphocyte function, and sensitization of antibody-producing cells to immune-suppressive drugs after plasma exchange.

INDICATIONS FOR PLASMA EXCHANGE s0025

Evidence to support specific indications for plasma exchange is variable in quality. Direct comparison among RCTs can be unsatisfactory because of variations in dose and frequency of plasma exchange and in immunosuppressive and other adjunctive therapy. The American Society for Apheresis (ASFA) reviewed all indications for plasma p0070 exchange most recently updated in 2016 and summarized available trial data.[7,8] In this chapter, evidence from available RCTs is discussed alongside observational data. The indications are summarized in Tables 104.3 and 104.4.

s0030 Anti–Glomerular Basement Membrane Antibody Disease (Goodpasture Disease)

p0075 Most patients can be depleted of pathogenic anti-GBM antibodies after 7 to 10 plasma volume exchanges if further antibody synthesis is inhibited by the concurrent use of cyclophosphamide and corticosteroids. Before plasmapheresis, the mortality from Goodpasture disease was higher than 90%, and only 11% of patients who were not dialysis-dependent at presentation survived with preserved kidney function. The use of plasma exchange improved the outcome considerably: 70% to 90% of patients now survive. However, only 50% of survivors retain independent kidney function and 10% or less of those who are dialysis dependent at presentation. There has been only one small controlled trial of plasma exchange in the treatment of Goodpasture disease, which used a low intensity of plasma exchange.[7,9] A total of 17 patients were randomized to receive corticosteroids and cyclophosphamide, with or without plasma exchange. Only 2 of the 8 who received plasma exchange developed end-stage kidney disease (ESKD) compared with 6 of the 9 who received drugs alone.

Long-term data from 71 patients with Goodpasture disease con- p0080
firmed the benefit of a treatment regimen including plasma exchange because most patients with mild to moderate kidney failure retained independent kidney function over 10 to 25 years,[10] and kidney recovery was possible even in some of those with the most severe kidney disease. Very similar results were identified in the largest reported series from China. Combining all the available published data for patients with Goodpasture disease, 76% of patients presenting with serum creatinine less than 5.5 to 6.8 mg/dL (500–600 μmol/L) will remain independent of dialysis if treated with plasma exchange, in contrast to only 8% of those who are dialysis dependent at presentation. Diffuse alveolar hemorrhage, which occurs in up to 50% of patients and can be life threatening, is an independent indication for plasma exchange regardless of kidney function.

Recommendation s0035

All patients who are not dialysis dependent at presentation should p0085
receive intensive plasma exchange with daily 4-L exchanges initially for 14 days (regimen shown in Table 104.2). For dialysis-dependent patients, we recommend plasma exchange with immunosuppression only for those who have biopsy or clinical evidence of recent-onset disease. Pulmonary hemorrhage is an independent indication for plasma exchange. Treatment of Goodpasture disease is discussed further in Chapter 25.

t0020 **TABLE 104.3 Conditions for Which There Is Strong Evidence for the Benefit of Plasma Exchange**

Indication	RCTs (No. Patients)	Controlled Trials (No. Patients)	Case Series (No. Patients)	Replacement Fluid	Comments
ANCA-associated systemic vasculitis	9 (652)	1 (26)	>22 (>347)	Albumin unless pulmonary hemorrhage or need to prevent coagulopathy	Possible benefit only in dialysis-dependent patients or those with most severe kidney impairment. Should consider daily exchanges in fulminant cases or with pulmonary hemorrhage.
Anti-GBM antibody disease	1 (17)	0	>10 (>587)	Albumin unless pulmonary hemorrhage or need to prevent coagulopathy	Minimum course 14 days to remove antibodies effectively. Especially beneficial in nonoliguric patients predialysis. Patients with creatinine >5.5 mg/dL (500 μmol/L) unlikely to benefit.
Cryoglobulinemia	1 (57) 1 (17) using immunoadsorption	0	24 (302)	Albumin	Long-term maintenance treatment needed in some patients. Ensure blood warmer on return lines or warm replacement fluids.
TTP	7 (301)	2 (133)	>38 (>1541)	Plasma or cryo-poor plasma	Daily, often with corticosteroids; the only treatment that has improved mortality.
ABO-incompatible kidney transplantation	0	0	>21 (>750)	Albumin ± plasma (compatible with donor and recipient or AB)	Used pretransplant to reduce titers of antibodies and often continued for a few days after surgery to allow successful transplantation.
Antibody-mediated kidney transplant rejection	3 (61)	8 (342)	>37 (>727)	Albumin	Daily or alternate day. Usually with IVIG and sometimes enhanced immunosuppression.
HLA desensitization for transplantation (in highly sensitized patients)	0	5 (441)	29 (466)	Albumin	Always in combination with immunosuppression and usually IVIG, and continued until cross-match negative. Usually five plasma exchanges are needed to reduce Ab levels sufficiently.

ANCA, Antineutrophil cytoplasmic antibody; *GBM*, glomerular basement membrane; *HLA*, human leukocyte antigen; *IVIG*, intravenous immunoglobulin; *RCTs*, randomized controlled trials; *TTP*, thrombotic thrombocytopenic purpura.

t0025 **TABLE 104.4 Conditions for Which There Is Some Evidence for Plasma Exchange**

Indication	RCTs (No. of Patients)	Controlled Trials (No. of Patients)	Case Series (No. of Patients)	Replacement Fluid	Comments
Catastrophic antiphospholipid antibody syndrome	0	0	6 (109)	Plasma	Should be done daily. Combination of plasma exchange or IVIG, heparin, and corticosteroids (from registry data) gives best outcomes.
Recurrent FSGS after transplantation	0	3 (48)	77 (378)	Albumin	Sometimes in combination with rituximab. May need long-term maintenance treatment.
Atypical hemolytic-uremic syndrome	0	0	>10 (>200)	Plasma or cryo-poor plasma	Daily plasma exchange initially. Eculizumab is the preferred treatment if available.
Myeloma	5 (182)	0	8 (102)	Albumin	Daily or alternate day for 7–10 exchanges. Despite negative randomized trial in 2005, plasma exchange might be considered if high light-chain load, severe kidney failure, and oliguria and light chains not depleted with urgent chemotherapy.
Rapidly progressive glomerulonephritis (may include patients with ANCA-associated disease in older literature)	7 (196)	0	21 (295)	Albumin	No good evidence for benefit in immune complex disease of any cause. Small case series of benefit in crescentic IgA vasculitis (Henoch-Schonlein purpura) with RPGN.
Scleroderma	0	3 (75)	7 (70)	Albumin	No good evidence for benefit, but some patients have reported improvement.
Systemic lupus (not nephritis)	1 (20)	1 (4)	14 (128)	Albumin	For cerebritis, lupus-associated TTP, severe hemolysis, or pulmonary hemorrhage.

ANCA, Antineutrophil cytoplasmic antibody; *FSGS,* focal segmental glomerulosclerosis; *IgA,* immunoglobulin A; *IVIG,* intravenous immunoglobulin; *RCTs,* randomized controlled trials; *RPGN,* rapidly progressive glomerulonephritis; *TTP,* thrombotic thrombocytopenic purpura.

s0040 ## Small-Vessel Vasculitis

p0090 The majority of patients with rapidly progressive glomerulonephritis (RPGN), other than anti-GBM disease, have small-vessel vasculitis with ANCA detectable in their serum, and there is increasing evidence that these autoantibodies are pathogenic. Several trials of plasma exchange in non–anti-GBM RPGN have been reported.[7,11] Most of the early trials included patients with a variety of diseases, used a low intensity of plasma exchange, and often excluded those with oligoanuria. These trials showed no overall benefit of plasma exchange in addition to conventional immunosuppression; however, those patients with the most severe disease did seem to benefit. Two recent large RCTs have, however, added to the controversy. MEPEX (Methylprednisolone or Plasma Exchange in Severe ANCA-Associated Vasculitis) randomized 137 patients with ANCA-associated systemic vasculitis and serum creatinine greater than 5.5 mg/dL (500 μmol/L) to plasma exchange or IV methylprednisolone in addition to oral corticosteroids and cyclophosphamide.[11] Sixty-nine percent of patients recovered kidney function when treated with plasma exchange compared with 49% of those receiving IV methylprednisolone, and significantly more patients were dialysis independent at the trial endpoint, although this difference was not maintained at 3 years. MEPEX was the largest study in a meta-analysis of 387 patients, with creatinine levels ranging from 3.2 to 13.5 mg/dL. The addition of plasma exchange to standard immunosuppression in MEPEX was associated with reduced risk for ESKD or death. The PEXIVAS trial randomized 352 patients with ANCA-associated vasculitis and estimated glomerular filtration rate (eGFR) less than 50 mL/min to plasma exchange (7 exchanges in 14 days) or no plasma exchange, and also low- or high-dose oral steroids.[12] Overall, there was no difference in deaths or ESKD with or without plasma exchange (28.5% vs. 31%; hazard ratio [HR] 0.86, 95% confidence interval [CI] 0.65–1.13) but a suggestion that plasma exchange might increase the chance of kidney recovery in those with the most severe kidney impairment at presentation, and a new meta-analysis including PEXIVAS is awaited.

Patients with both ANCA and anti-GBM antibodies (so-called double-positive patients) and RPGN do not respond well to plasma exchange and rarely, if ever, recover kidney function.[13] p0095

Recommendation s0045

We perform plasma exchange in patients with small-vessel vasculitis who present with severe kidney failure (serum creatinine > 5.5 mg/dL [~500 μmol/L] or dialysis dependent) or pulmonary hemorrhage. The regimen is shown in Table 104.2. p0100

Other Crescentic Glomerulonephritis s0050

Crescent formation is a common histologic finding in several other patterns of GN, including postinfectious GN, GN associated with infective endocarditis, IgA nephropathy (IgAN), membranoproliferative glomerulonephritis (MPGN), and membranous nephropathy. Such patients were often included in studies of treatment of RPGN, and plasma exchange has been used in the treatment of a number of these conditions. More than 400 patients with such diseases have been treated with plasma exchange with no good evidence for any benefit in crescentic GN not caused by anti-GBM disease or vasculitis.[7,8] In crescentic IgAN, there are anecdotal reports of short-term benefit in patients with severe kidney impairment, but longer-term follow-up has proved disappointing. A single report showed some benefit of 5 to 10 plasma exchanges in preventing or reversing dialysis dependency in 12 patients with crescentic IgAN manifesting with RPGN, with measured decreases in circulating plasma IgA-IgG complexes.[14] p0105

s0055 Recommendation

p0110 We offer plasma exchange in patients with IgAN and other GNs who have rapidly deteriorating kidney function and extensive fresh crescents in the biopsy specimen.

s0060 Focal Segmental Glomerulosclerosis

p0115 Plasma exchange, protein A immunoadsorption, and lipoprotein apheresis have been used to treat patients with primary FSGS or recurrent disease after transplantation. The results have been worse in primary disease with less than 40% of patients achieving either partial or complete remission,[7,8,13] and we do not recommend plasma exchange in this setting. Plasma exchange for recurrent disease is discussed later and in Chapter 113.

s0065 Thrombotic Microangiopathies

p0120 In both HUS and TTP, endothelial activation leads to TMA but through distinct mechanisms.

s0070 Infection-Associated Hemolytic-Uremic Syndrome

p0125 Infections leading to HUS are most recognized after enteropathogenic *Escherichia coli* from Shiga toxin production (Stx-associated HUS) and after *Streptococcus pneumoniae* (pHUS). The prognosis is generally good, especially in childhood. Most children will recover extremely well with supportive care and management of fluid and electrolyte imbalance and hypertension. Two controlled trials of plasma infusion (at least 10 mL/kg/day) in childhood HUS complicated by dialysis-dependent kidney failure showed no clinical benefit (as determined by hypertension, kidney dysfunction, and proteinuria) in either short- or medium-term follow-up.[15] There has been no study of plasma exchange in childhood Stx-associated HUS. Plasma exchange and infusion have not been subjected to any controlled trials in adult Stx-associated HUS, but uncontrolled observations suggest possible benefit.[7,16] The 2011 outbreak of enterohemorrhagic and Stx-producing *E. coli* O104:H4 in Europe led to 855 confirmed cases of HUS, but despite severe illness in many patients, a retrospective analysis of 491 treated patients did not support any major benefit of plasma exchange in addition to intensive supportive care.[17]

s0075 Thrombotic Thrombocytopenic Purpura

p0130 Patients with TTP usually have a defective vWF cleaving protease (ADAMTS13), an enzyme that normally degrades large vWF multimers. The defect is typically because of an inherited deficiency or autoantibodies directed against the protease. Accumulation of vWF multimers leads to systemic platelet activation under conditions of high shear stress (the microcirculation) and thrombosis. The rationale for plasma infusion and plasma exchange in TTP is therefore to replenish vWF cleaving protease, to remove antibodies against the protease, and to remove the large vWF multimers from circulation. There are well-designed RCTs in the treatment of TTP.

p0135 The first prospective controlled trial compared plasma infusion with plasma exchange (1–1.5 plasma volumes at least seven times in the first 9 days).[18] Of patients receiving plasma exchange, 47% had a platelet count exceeding 150×10^9 cells/L and no new neurologic features, compared with only 25% of those receiving plasma infusion over the first 2 weeks. At 6 months, survival was substantially better in those given plasma exchange (50% vs. 78%). More recent series using plasma exchange have reported mortality rates as low as 15%,[7] and there may be an association of reduced early mortality with more intensive plasma exchange. Kidney impairment is not an independent predictor of poor outcome in TTP and does not in itself warrant more intensive therapy.

TTP may also be induced by drugs, including ticlopidine, clopidogrel, mitomycin C, cyclosporine, tacrolimus, gemcitabine, and quinine, and the evidence for benefit of plasma exchange in this context is poor, except for ticlopidine-induced TTP in which ADAMTS13 activity is severely depressed.[7] p0140

Atypical Hemolytic Uremic Syndrome s0080

The less common forms of HUS in which there is no clear diarrheal prodrome (atypical HUS [aHUS]) are commonly caused by mutations, polymorphisms, or acquired dysregulation of the complement pathway (including the development of autoantibodies), especially of factor H, factor I, and membrane cofactor protein, leading to uninhibited activation of complement. Other causes include infections or drugs that cause platelet or leukocyte activation and complement activation and consumption. Direct activation of endothelial cells also may be a cause. Plasma exchange and infusion have not been evaluated in controlled trials in aHUS, but uncontrolled series suggest benefit and current guidelines recommend early initiation of plasma exchange with FFP (see Chapter 30), both to remove potential complement inhibitors or autoantibody and to replace absent or defective complement regulators.[7] The introduction of the complement inhibitor eculizumab, a humanized anti-C5 monoclonal antibody, has revolutionized the treatment of aHUS and should now be the mainstay of management after urgently excluding other diagnoses and can replace plasma exchange. If eculizumab is not available, plasma exchange should be continued. TMA after kidney transplantation has also responded to plasma exchange. p0145

Recommendation s0085

We use plasma exchange in all adults with TTP and perform all exchanges using FFP or cryo-poor FFP as the exchange fluid. In aHUS, we use plasma exchange until we can initiate eculizumab, which is the preferred treatment. p0150

Systemic Lupus s0090

Plasma exchange has been used extensively in patients with lupus. Most studies have included patients with diverse patterns of disease, often with only mild kidney involvement. A prospective RCT could show no benefit of plasma exchange over conventional immunosuppression for kidney, serologic, or clinical outcomes, both in the short and long term.[19] However, patients with crescentic lupus nephritis and those with the most severe kidney dysfunction (dialysis dependency) were excluded. Anecdotal evidence suggests that plasma exchange may benefit patients with systemic lupus and crescentic GN, pulmonary hemorrhage, cerebral lupus, catastrophic antiphospholipid syndrome, severe antibody-induced hemolysis, lupus-associated TTP, or severe lupus unresponsive to conventional drugs or in patients for whom cytotoxic therapy has been withdrawn because of bone marrow suppression or other toxicity. Immunoadsorption may be more successful in the severe forms of lupus nephritis. A variety of techniques have been used, including standard protein A and anti-immunoglobulin absorption, and also phenylalanine, tryptophan, and dextran sulfate ligands, all of which bind immunoglobulin, rheumatoid factors, and immune complexes to varying degrees and all of which have been reported to induce remission in patients with severe disease after failure of conventional therapy. p0155

Recommendation s0095

We offer plasma exchange to lupus patients with rapidly progressive kidney failure and class IV lupus nephritis with crescents, to patients with severe neurologic involvement or severe hemolysis, to patients with myelosuppression who are thought unable to tolerate p0160 cyclophosphamide, and to those with catastrophic antiphospholipid syndrome or severe macrophage activation syndrome. The treatment of lupus is further discussed in Chapter 27.

s0100 Cryoglobulinemia

p0165 In type I cryoglobulinemia, usually associated with myeloma or lymphoma, a monoclonal immunoglobulin causes hyperviscosity and cryoprecipitation. Such antibodies are easily removed by plasma exchange, often with immediate clinical benefit. Cytotoxic agents are used simultaneously to inhibit further paraprotein production. There are no controlled trials of plasma exchange, but symptoms are closely related to the presence of the cryoimmunoglobulin, and hence treatment with plasma exchange appears effective.[7]

p0170 Patients with type II (mixed essential) cryoglobulinemia develop a monoclonal antibody (usually IgM) with specificity for a second, usually polyclonal, immunoglobulin. Type II cryoglobulins occur most commonly in association with hepatitis C virus (HCV) infection and lymphoma. The resulting immune complexes can be deposited in the microcirculation and are particularly associated with an MPGN pattern of injury (see Chapter 22). Plasma exchange is effective at clearing the immune complexes, although the cryoglobulins often recur, and sustained benefit has not been clearly demonstrated. However, many of the acute features of cryoglobulinemia resolve with plasma exchange, particularly arthralgia, skin lesions, and digital necrosis, and patients with RPGN can recover kidney function. Concomitant treatment with rituximab may prevent resynthesis of the cryoproteins, although some patients require long-term intermittent plasma exchange to control symptoms. Patients with HCV-associated cryoglobulinemia should be treated with the newer antiviral agents, which are extremely effective in curing the viral infection, after which cryoglobulins usually disappear. A single RCT in 17 patients with HCV-associated cryoglobulinemia added immunoadsorption apheresis (with dextran sulfate) to antivirals and immunosuppression and showed significant clinical improvements, but this was from an era before the current antiviral agents.[20]

p0175 Cryofiltration apheresis (in which a normal plasma filter is used to separate plasma, which is then cooled to precipitate the cryoglobulin before return to the patient) selectively removes cryoglobulins, avoids large volumes of replacement fluids, and avoids deficiency of clotting factors, but needs to be combined with immunosuppression to prevent synthesis of further cryoglobulin. Few centers currently perform this technique, especially because of the widespread introduction of rituximab for cryoglobulinemia.

s0105 Recommendation

p0180 We offer plasma exchange (in addition to rituximab or antiviral treatments) to patients with cryoglobulinemia and rapidly progressive kidney failure, necrotic ulceration, and digital infarction and to patients with type 1 cryoglobulins and complications from hyperviscosity.

s0110 Myeloma

p0185 Plasma exchange almost certainly does not provide benefit in myeloma with either cast nephropathy or light-chain kidney toxicity, and the most important therapy seems to be urgent initiation of chemotherapy, especially thalidomide, lenalidomide, bortezomib, or newer agents. A large prospective RCT randomized 97 patients with myeloma and progressive acute kidney injury (creatinine > 200 μmol/L [2.3 mg/dL] with an increase of over 50 μmol/L over the previous 2 weeks despite conventional management) to receive plasma exchange (five to seven sessions of 50 mL/kg over 10 days) in addition to chemotherapy (vincristine, adriamycin, and dexamethasone [VAD] or melphalan and prednisolone).[21] This study showed no benefit of plasma exchange on mortality or recovery of kidney function. However, patients had a wide degree of kidney dysfunction, and relatively few had a kidney biopsy performed to confirm cast nephropathy. A retrospective review suggested that those with myeloma and high light-chain loads or severe kidney failure may benefit if plasma exchange reduces light chains rapidly.[22] More recently a variety of studies in patients with kidney disease and myeloma, especially using bortezomib-based regimens, have shown significant improvements even in patients with dialysis-requiring kidney failure but without plasma exchange. An RCT of HD using novel membranes allowing the removal of large-molecular-weight substances and lengthy dialysis sessions (6–8 hours) failed to show any benefit (for survival or recovery of kidney function) when added to modern chemotherapy, which depleted light chains rapidly in the absence of plasma removal.

Recommendation s0115

We no longer use plasma exchange in patients with myeloma but p0190
ensure that effective chemotherapy is urgently initiated.

TRANSPLANTATION s0120

Antibody-Mediated Rejection s0125

A review of 157 patients included in five trials did not show any bene- p0195
fit of plasma exchange for the treatment of acute vascular rejection.[7,8] At least 11 trials including more than 400 patients with more clearly defined antibody-mediated rejection and case series of more than 700 patients have suggested that plasma exchange, combined with IV immunoglobulin (IVIG) and/or rituximab, but sometimes antithymocyte globulin, may reverse 55% to 100% of such rejection episodes.[7]

There is no convincing evidence that plasma exchange has any role p0200
in the treatment of chronic rejection.

Anti–Human Leukocyte Antigen Antibodies s0130

Highly sensitized patients with preformed anti-HLA antibodies have p0205
been treated before and after transplantation with plasma exchange or immunoadsorption to reduce cytotoxic antibody levels, often with high-dose IVIG.[7] Patients usually received intensive immunoadsorption or plasma exchange before transplantation to ensure a current negative crossmatch immediately before transplantation; some received longer-term immunoadsorption or plasma exchange in combination with immunosuppressive therapies in the months before transplantation. Most recent studies have shown that donor-specific antibody titers of less than 1 to 32 are often depleted completely with preoperative plasma exchange, allowing successful kidney transplantation. Such patients have an increased risk for antibody-mediated rejection—approximately 40%—but despite this, 90% have 1-year graft survival.

ABO-Incompatible Kidney Transplantation s0135

Plasma exchange is widely used to remove natural anti-A or anti-B p0210
blood group antibodies from the recipient before living donor transplantation from an ABO-incompatible donor. Various protocols are in use, but all rely on depletion of specific antibody over 2 to 5 days before transplantation by exchanging a single plasma volume for human albumin solution (in addition to routine immunosuppression, sometimes including rituximab and IVIG). Plasma exchange is sometimes continued for one or two sessions after transplantation or if antibody-mediated rejection occurs.[23] One-year graft survival rates of over 90% have been reported with such protocols, and although rejection episodes are more common than in ABO-compatible transplants, overall graft survival is similar. Subsequent return of ABO antibodies in the weeks after the kidney transplant does not usually cause rejection (a process termed accommodation and which is not fully explained).

Patients with increasingly high antibody titers are being treated in this way. Immunoadsorption using synthetic A- or B-oligosaccharide epitopes linked to Sepharose has been developed, which specifically remove anti-A or anti-B antibodies, but any clinical benefit remains uncertain and the costs are high.

s0140 ## Recurrent Focal Segmental Glomerulosclerosis

p0215 Plasma exchange, double-filtration plasma exchange, and protein A immunoadsorption have been used to treat nephrotic syndrome after transplantation in patients with recurrent FSGS.[7,24,25] An incompletely defined circulating factor causing increased permeability of glomerular capillaries can be found in most patients with recurrent FSGS. There are no controlled trials of plasma treatments in recurrent FSGS, and most series are small. A meta-analysis and systematic review of 77 case reports including 378 patients demonstrated an overall remission rate of 71%, with 46.8% achieving a complete remission after plasma exchange.[25] The study suggested that those with a more rapid relapse after transplantation and lower proteinuria were more likely to respond, with no difference between deceased and living donors. Patients received a median of 12 plasma exchange sessions. All three apheresis modalities have been used prophylactically in patients deemed to be at high risk for recurrence, with variable success. Intriguingly, lipoprotein apheresis (LDL-A) has been shown in small case series to induce remission of proteinuria in both primary and recurrent FSGS, via unknown mechanisms. LDL absorption occurs using dextran sulfate cellulose columns which absorb and remove LDL, VLDL, and lipoprotein (a) from plasma. Treatments are undertaken weekly for 6 to 12 weeks, and in very small and selected series achieve remission rates of approximately 50%.[26]

Recommendations s0145

We recommend plasma exchange for patients with recurrent FSGS, p0220
started early, initially daily for 7 to 10 days. If proteinuria is successfully reversed, this may need to be continued less frequently for several months. Management of recurrent FSGS is discussed further in Chapter 113 We recommend plasma exchange for acute antibody-mediated rejection together with enhanced immunosuppression.

SELF-ASSESSMENT QUESTIONS

o0010 1. A patient with anti–GBM disease, serum creatinine 250 µmol/L, and pulmonary hemorrhage is being treated with plasma exchange and immunosuppression. Which complication can occur as a direct result of the plasma exchange?
o0015 **A.** Hypernatremia
o0020 **B.** Hyperviscosity
o0025 **C.** Hypocalcemia
o0030 **D.** Pulmonary embolism
o0035 **E.** Thrombocytosis

o0040 2. A 52-year-old woman presented with acute kidney injury and evidence of hemolysis and thrombocytopenia. The results of investigations showed:
u0010 • Hemoglobin: 86 g/L (115–165)
u0015 • White blood cell count: 4.2×10^9/L (4.0–11.0)
u0020 • Platelet count: 35×10^9/L (150–400)
u0025 • Serum creatinine: 238 µmol/L (60–110)

The results of remaining investigations are awaited. Which one of u0030
these diagnoses should lead to urgent initiation of plasma exchange?
A. Atypical HUS o0045
B. Infection-associated HUS o0050
C. Myeloma o0055
D. Scleroderma renal crisis o0060
E. Systemic lupus o0065

3. A 42-year-old man received a cadaveric kidney transplant under o0070
standard immunosuppression. Six weeks later, his kidney function deteriorated, an ultrasound scan was normal, and a kidney biopsy was performed. Which finding might lead to initiation of plasma exchange?
A. Antibody-mediated rejection o0075
B. $CD4^+$ staining on the biopsy o0080
C. Cyclosporine-induced microangiopathy o0085
D. Recurrent membranous nephropathy o0090
E. T-cell–mediated cellular rejection o0095

s0150 # REFERENCES

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